



There is not enough spectrum in general and 5G is a lot about optimising the use of spectrum.

New mobile generations are typically assigned new frequency bands and wider spectral bandwidth per frequency channel (1G up to 30kHz, 2G up to 200kHz, 3G up to 20MHz and 4G up to 100MHz), but skeptics argue that there is little room for

schemes, it will be possible to link to others nearby to provide ad-hoc wireless networks for much speedier data flows.

Smart antennae. Another major element of any 5G mobile system will be that of smart antennae. Using these, it will be possible to alter the beam direction to enable more direct communication, limit interference and increase overall cell capacity.

Scepticism about 5G networks

Every new mobile standard brings with it calls from operators for more spectrum, and 5G is no exception. If mobile operators want to deliver more and more capacity, they will need more and more wireless spectrum to do so. And, with every generation of mobile tech, governments around the world must identify what spectrum those operators will need, whether anyone is using those bands and how to move them off those, if so. And then find the best way to sell that spectrum at the right price and finally make sure that all operators are meeting the obligations that buying the spectrum imposed on them.

The history of the wireless industry is littered with tales of fouled-up spectrum auction procedures, delays to network rollouts, mud-slinging between mobile companies, obligations not met and clean up procedures not followed. Spectrum is and will remain a major challenge for the success and early rollout of 5G.

larger channel bandwidths and new frequency bands suitable for land-mobile radio.

From users' point of view, previous mobile generations have implied substantial increase in peak bit rate (that is, physical layer net bit rates for short-distance communication), up to 1Gbps, to be offered by 4G. If 5G appears, it reflects these prognoses.

The major issue with 5G technology is that there is such an enormously wide variation in requirements—superfast downloads to small data requirements for the IoT—that any one system will not be able to meet these needs. Accordingly, a layer approach is likely to be adopted. It is rightly said that 5G is not just a mobile technology but an ubiquitous access to high and low data rate services.

Challenges associated with next-generation network

There are many new techniques and technologies that will be used in the new 5G mobile telecommunications system. These new 5G technologies are still being developed and overall standards have not yet been defined. However, as required technologies develop, these will be incorporated into the new system, which will be defined by the standards bodies over the coming years.

There are several key areas that are being investigated by research organisations. These include:

SINCE 1962

LOW COST LCD / LED MODULES

LC035

LC135

Specifications

3½ Digits, 1999 Counts (max.) LCD	3½ Digits, 1999 Counts (max.) LED
LC035	LC135
0 - 200mV DC 0 - 2, 20, 200V DC 0 - 200µA DC 0 - 2, 20, 200mA DC	
Auxiliary Power Supply	
9VDC ± 10%	5VDC ± 10%

Applications

• Ammeter	• Watt Meter
• Voltmeter	• LUX Meter
• Thermometer	• LCR Meter
• Power Supply	• Capacitance Meter
• PH Meter	• Other Industrial and Domestic uses.
• DB Meter	

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